

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:26-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

--	--	--	--	--	--

26-01-24_SR.*CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-27(SPCL)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

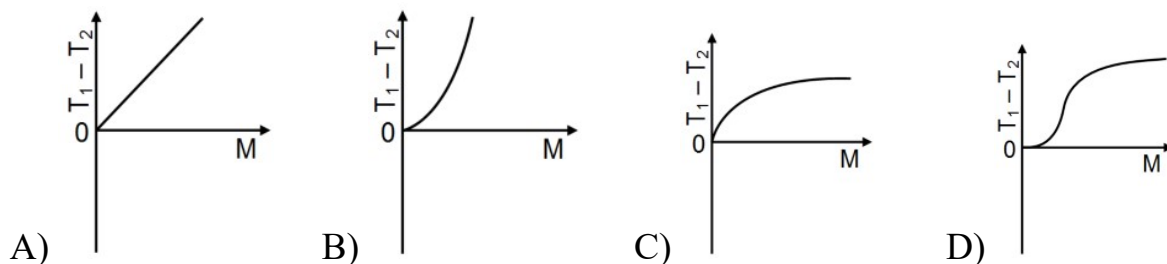
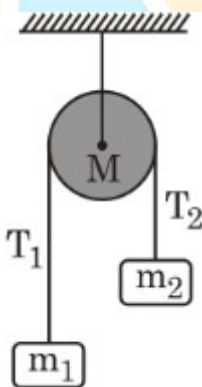
For More Material Join: @JEEAdvanced_2024

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two uniform solid spheres A and B of same material, painted completely black and placed in free space separately where no radiation is incident on spheres. Their radii are R and $2R$ respectively and the dominating wavelengths (wavelength corresponding to which spectral emissive power is maximum) in their spectrum are observed to be in the ratio $1 : 2$. Which of the following is not correct?
- A) Ratio of their temperatures is $2 : 1$
- B) Ratio of their emissive powers is $4 : 1$
- C) Ratio of their rates of heat loss is $4 : 1$
- D) Ratio of their rates of cooling is $32 : 1$
2. On a pulley of mass M hangs a rope with two masses m_1 and m_2 ($m_1 > m_2$) tied at the ends as shown in the figure. The pulley rotates without any friction, whereas the friction between the rope and the pulley is large enough to prevent any slipping. Which of the following plots best represents the difference between the tensions in the rope on the two sides of the pulley as a function of the mass of the pulley?

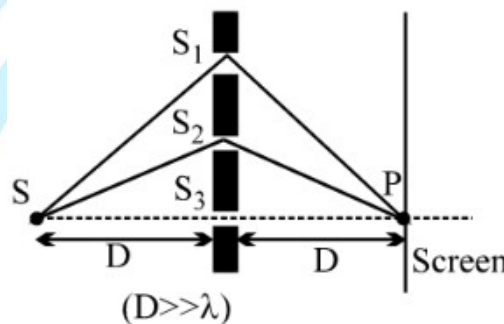


3. **Assertion :** If the earth expands in size without any change in mass, the length of the day would increase.

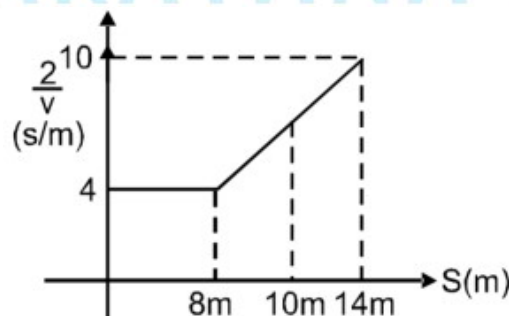
Reason : The rotation of earth about its axis follows the law of conservation of angular momentum.

- A) Assertion is true, Reason is true; Reason is the correct explanation of Assertion.
 B) Assertion is true, Reason is true; Reason is not the correct explanation of Assertion.
 C) Assertion is false, Reason is true.
 D) Assertion is true, Reason is false.
4. A monochromatic light source of wavelength λ is placed at S. Three slits S_1, S_2 and S_3 are equidistant from the source S and the point P on the screen.

Given that: $S_1P - S_2P = \frac{\lambda}{6}$ and $S_1P - S_3P = \frac{2\lambda}{3}$. If I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is:

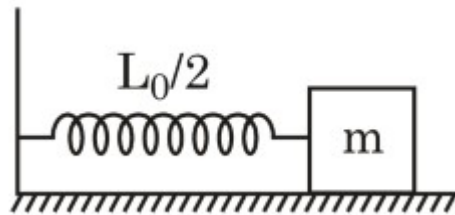


- A) 3 I B) 5 I C) 8 I D) Zero
5. Figure shows $\frac{2}{v} v/s$ curve for a particle of mass 2 kg moving in a straight line. If the time (in seconds) taken by the particle to achieve a displacement of 10 m from start is t. (v = velocity, s = displacement), then find the value of $(t - 20)$.



- A) 1 s B) 21 s C) 3 s D) 42 s

6. If the surface of a metal is successively exposed to radiation of $\lambda_1 = 350 \text{ nm}$ and $\lambda_2 = 450 \text{ nm}$, the maximum speed of photoelectrons is halved. The work function of this metal is closest to (given $h = 6.62 \times 10^{-34} \text{ J-s}$)
- A) 1.8 eV B) 2.5 eV C) 4.8 eV D) 3.9 eV
7. A block of mass m is pushed against a spring whose spring constant is k fixed at one end with a wall. The block can slide on a frictionless table as shown in figure. If the natural length of spring is L and it is compressed to half its length when the block is released, find the velocity of the block, when the spring has natural length.



- A) $\sqrt{\frac{m}{k} \cdot \frac{L_0}{2}}$ B) $\sqrt{\frac{k}{m} \cdot \frac{L_0}{2}}$ C) $\sqrt{\frac{k}{m}} L_0$ D) $\sqrt{\frac{kL_0}{m}}$
8. A block of ice of mass 20 gm is kept in a steel container of water equivalent 10gm. The temperature of both ice & the container is -30°C . Now 30 gm water at 80°C is poured in this container. Find the final common temperature ($S_{\text{ice}} = 0.5 \text{ cal/g}^\circ\text{C}$, $S_{\text{water}} = 1 \text{ cal/g}^\circ\text{C}$, $L_{\text{ice}} = 80 \text{ cal/g}$):-
- A) 0°C B) -5.55°C C) 3.33°C D) 5.55°C
9. Liquid is completely filled in a vessel up to height H and a hole is made at a depth h below the free surface of the liquid. Given that area of container is A_0 and area of orifice is A . Match the entries of List-1 with the entries in List-2.

	LIST- 1		LIST-2
P)	Velocity of efflux	1)	Proportional to $\sqrt{Hh - h^2}$
Q)	The range of liquid coming out of the orifice & filling on the ground	2)	Proportional to A_0
R)	If the hole is at the bottom of the tank, time t taken to empty the tank is	3)	Proportional to $\sqrt{h}$

S)	Horizontal thrust force on the system (Liquid + Container) due to the liquid flowing out of container	4)	Proportional to $\sqrt{H}$
----	---	----	----------------------------

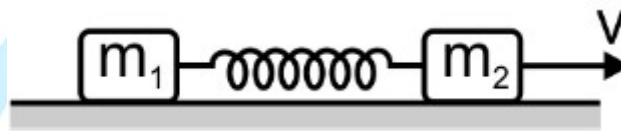
A) (P)→(1), (Q)→(3), (R)→(4), (S)→(2)

B) (P)→(2), (Q)→(1), (R)→(4), (S)→(3)

C) (P)→(3), (Q)→(1), (R)→(4), (S)→(2)

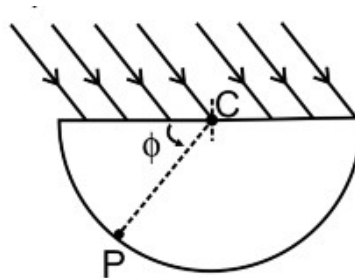
D) (P)→(3), (Q)→(2), (R)→(4), (S)→(1)

10. Two blocks of mass m_1 and m_2 ($m_1 < m_2$) are connected with an ideal spring on a smooth horizontal surface as shown in figure. At $t = 0$, m_1 is at rest and m_2 is given a velocity v towards right. At this moment, spring is in its natural length. Then choose the correct alternative:



- A) Block of mass m_2 will be finally at rest after some time.
 B) Block of mass m_2 will never come to rest.
 C) Both the blocks will be finally at rest.
 D) None of these
11. Parallel light rays are falling on the plane surface of semi cylindrical rod made of glass of refractive index 2.0 at an angle of incident of 60° in such a plane which is perpendicular to the axis of semi-cylinder as shown in figure. If ϕ is position where the rays are emerging out of cylindrical surface after refraction then ϕ lies between :

[Given: $\sin^{-1} \frac{\sqrt{3}}{4} = 25^\circ$]



- A) $85^\circ < \phi < 145^\circ$ B) $145^\circ < \phi < 180^\circ$ C) $75^\circ < \phi < 105^\circ$ D) $25^\circ < \phi < 90^\circ$

12. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v , from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be :-

A) $(2f, 2f)$ B) $\left(\frac{f}{2}, \frac{f}{2}\right)$ C) (f, f) D) $(4f, 4f)$

13. A non-relativistic particle of mass m is held in a circular orbit around the origin by an attractive force $F(r) = -kr$, where k is positive constant. Assuming the Bohr quantization of the angular momentum of the particle, Select the correct alternative, if the radius of the orbit is r and speed of the particle is v .

A) $v^2 = \left(\frac{n\hbar}{m}\right)\left(\frac{k}{m}\right)^{1/2}$ B) $r^2 = \left(\frac{n\hbar}{k}\right)\left(\frac{k}{m}\right)^{1/2}$
C) $v^2 = \left(\frac{n\hbar}{2m}\right)\left(\frac{k}{m}\right)^{1/2}$ D) $r^2 = \left(\frac{n\hbar}{2k}\right)\left(\frac{k}{m}\right)^{1/2}$

14. A plane electromagnetic wave of wavelength λ has an intensity I . It is propagating along the positive Y-direction. The allowed expressions for the electric and magnetic fields are given by :-

A) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = +\frac{1}{c} E \hat{i}$

B) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = +\frac{1}{c} E \hat{i}$

C) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = -\frac{1}{c} E \hat{i}$

D) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = \frac{1}{c} E \hat{k}$

15. **Statement-1** : A uniformly charged disc has a pin hole at its centre. The electric field at the centre of the disc is zero.

Statement-2 : Disc can be supposed to be made up of many rings. Also electric field at the centre of uniformly charged ring is zero.

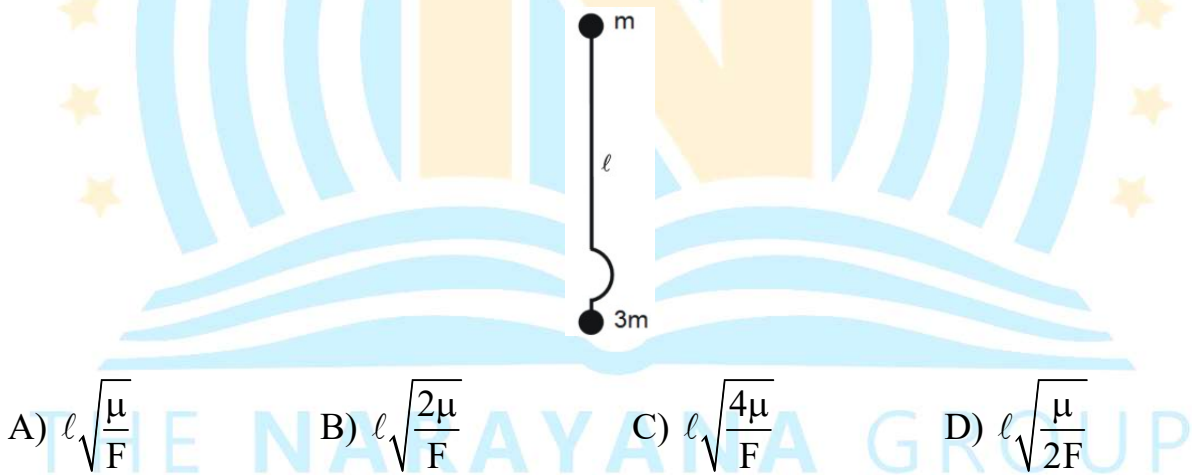
A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

C) Statement-1 is True, Statement-2 is False

D) Statement-1 is False, Statement-2 is True.

16. Two particles of mass m and $3m$ tied together with a very light string of mass density μ are dropped from a high rise tower as shown. A constant air friction force F acts on both of them. If the length of string is ℓ , the time taken by a pulse to go from bottom to top of string will be

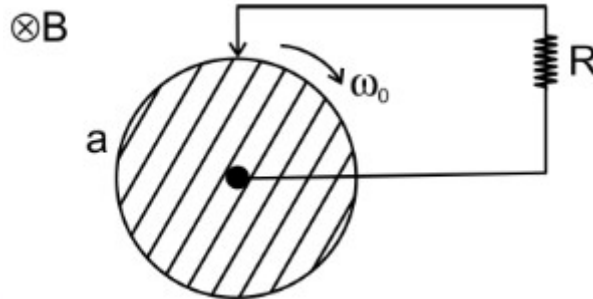


A) $\ell \sqrt{\frac{\mu}{F}}$ B) $\ell \sqrt{\frac{2\mu}{F}}$ C) $\ell \sqrt{\frac{4\mu}{F}}$ D) $\ell \sqrt{\frac{\mu}{2F}}$

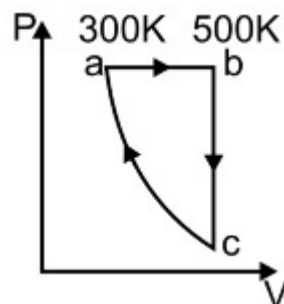
17. Internal resistance of battery of emf ε is given by $r = ki$, where k is positive constant and i is current drawn from battery. Maximum output power withdrawn from battery by a load resistor is –

A) $\frac{\varepsilon}{3} \sqrt{\frac{\varepsilon}{3k}}$ B) $\frac{2\varepsilon}{3} \sqrt{\frac{\varepsilon}{3k}}$ C) $\frac{\varepsilon}{3} \sqrt{\frac{2\varepsilon}{3k}}$ D) $\frac{2\varepsilon}{3} \sqrt{\frac{\varepsilon}{k}}$

18. A uniform conducting circular disc of mass m and radius a is hinged at centre. A resistance R is connected between centre and circumference by a slider. A uniform magnetic field is applied perpendicular to plane of disc. Now disc is given initial angular speed ω_0 , then initial angular acceleration of disc is (neglect resistance of disc)



- A) $\frac{\omega_0 B^2 a^2}{2mR}$ B) $\frac{2\omega_0 B^2 a^2}{3mR}$ C) $\frac{\omega_0 B^2 a^2}{mR}$ D) $\frac{\omega_0 B^2 a^2}{4mR}$
19. Which of the following statements is correct?
- A) Plane mirror always forms virtual image.
- B) Convex mirror always forms diminished image.
- C) In a spherical mirror, object and its image are always on the same side of focus.
- D) None of the above
20. A heat engine performs the reversible cycle $abca$ with 9.0 moles of an ideal gas. Path ca is an adiabatic process. The temperatures at point a and b are 300 K and 500 K, respectively. The adiabatic constant of the gas is 1.60. In figure, the heat absorbed by the gas in path ab , in kJ, is:-



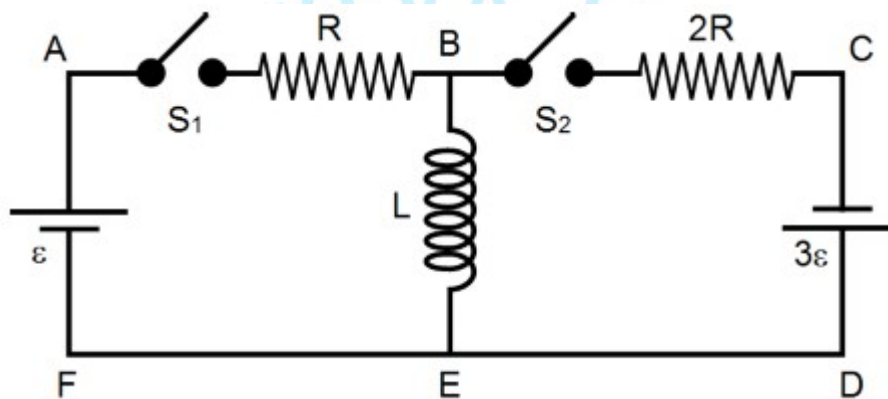
- A) 0 B) +25 C) +40 D) -25

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

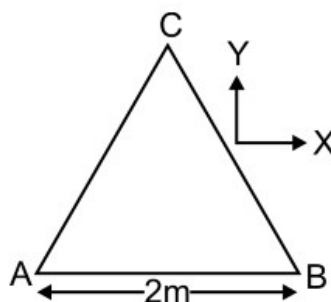
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

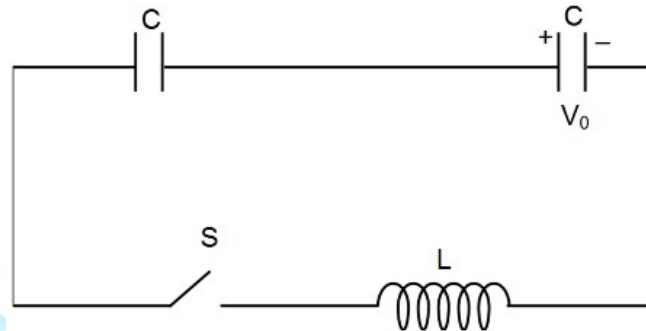
21. In the figure shown switch S_1 remains connected and switch S_2 remains open for a long time. Now S_2 is also closed. Assuming $\varepsilon = 10V$ and $L = 1H$. Find the magnitude of rate of change of current (in A/s) in inductor just after the switch S_2 is closed.



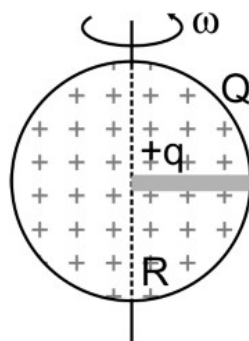
22. A non-conducting uniformly charged spherical shell is kept on a rough non-conducting horizontal surface. Mass of shell is 2kg and charge on it is 4C and friction coefficient between surface and shell is 0.18. Take $g = 10m/s^2$. The minimum value of external uniform horizontal electric field so that shell starts sliding on the surface is E_0 in SI units. Mark $100 E_0$ in answer.
23. A rigid equilateral triangular plate ABC of side 2 m, with side AB along the x-axis is in motion in the x-y plane as shown in figure. At the instant shown in the figure, the point B has velocity $\vec{v}_B = (3\hat{i} + 8\hat{j}) m/s$ and the plate has angular velocity $\vec{\omega} = 2\hat{k} rad/s$. Find the speed of point A (in m/s).



24. In the given electrical circuit both capacitors are identical. One of the capacitor is charged to voltage of V_0 volt. Now switch S is closed. Value of induced emf in inductor is $\frac{V_0}{\sqrt{a}}$ volt at the moment when energy in inductor is equal to half of total energy stored in capacitors. The value of a is.



25. In hydrogen atom, the binding energy of an electron in the ground state is E_1 , then the frequency of revolution of the electron in the n^{th} orbit is $\frac{x E_1}{n^y h}$. Here x and y are integers. Find the value of x^y .
26. A small ball of mass 1 kg and charge $\frac{2}{3} \mu\text{C}$ is placed at the centre of a uniformly charged sphere of radius 1 m and charge $\frac{1}{3} \text{mC}$. A narrow smooth groove is made in the sphere from centre to surface as shown in figure. The sphere is made to rotate about a diameter perpendicular to the groove at a constant rate of $\frac{1}{2\pi}$ revolutions per second. If the speed w.r.t. ground (in m/s) with which the ball slides out from the groove is $\frac{x}{16}$, then find x. (Neglect any magnetic effect)



27. A particle having charge 2 coulomb and mass $m = 8 \text{ Kg}$ is projected with initial velocity $\vec{v} = 3\hat{i} + 4\hat{j} \text{ m/s}$ in an electric field in which $\vec{E} = 5\hat{i} \text{ V/m}$. If initial radius of curvature of path of charged particle be 'R' then R (in m) is :-

(Take : $\vec{g} = -10\hat{j} \text{ m/s}^2$)

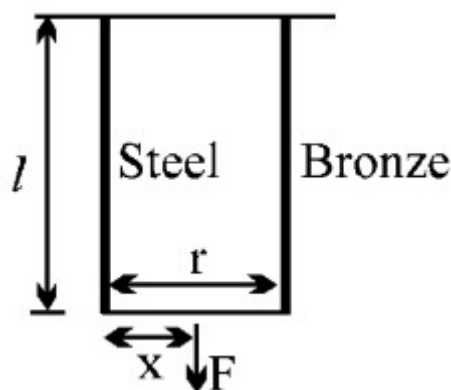
28. Height of a mountain at a place (h) is measured by measuring density of rock's material (d) forming the mountain, Young's Modulus of rock's material (Y) and acceleration due to gravity (g) at that place. Percentage error in measurement of d, Y and g are respectively $\pm 2\%$, $\pm 1\%$ and $\pm 2\%$, then find the magnitude of percentage error in measurement of h.

29. A 10 m long nichrome wire having 80Ω resistance has current carrying capacity of 10A. This wire can be cut into two equal parts and equal parts can be connected in series or parallel. The maximum power which can be obtained as heat by the wire from a 200 V main supply in (kW) is :-

30. Two vertical wires each $\ell = 150 \text{ cm}$ long one of steel having cross-sectional area $A_s = 2 \times 10^{-6} \text{ m}^2$ and second of bronze of cross-sectional area $A_b = 1 \times 10^{-6} \text{ m}^2$ are suspended from a ceiling with horizontal distance $r = 55 \text{ cm}$ as shown in figure. A light but rigid horizontal rod connect lower ends of wires. Where should a vertical downward force F be applied on the rod such that it remains horizontal even after application of force? Given :

for steel $Y_s = 2 \times 10^{11} \text{ N/m}^2$,

for bronze $Y_b = 1.5 \times 10^{11} \text{ N/m}^2$.

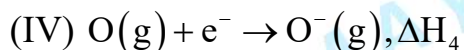
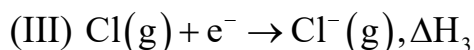
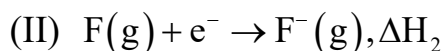
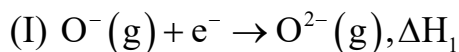


SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Consider the following reactions :



Select the INCORRECT statements

A) ΔH_4 is less negative than ΔH_2

B) ΔH_3 is more negative than ΔH_2 and ΔH_4

C) ΔH_3 and ΔH_4 are negative whereas ΔH_1 and ΔH_2 are positive

D) $\Delta H_2, \Delta H_3$ and ΔH_4 are negative whereas ΔH_1 is positive

32. A gaseous mixture of $H_2(g)$ and $O_2(g)$ has 70% $H_2(g)$ and 30% $O_2(g)$ by volume. If this gas mixture maintained at a total pressure of 2.5 atm is brought into contact with excess of liquid water at $20^\circ C$, then solution is found to dissolve 31.5 ml of H_2 per litre on saturation. Now pure hydrogen gas at a pressure of 1 atm is maintained in contact with liquid water at $20^\circ C$. Volume of H_2 per litre of solution dissolved will be :

A) 18 ml

B) 36 ml

C) 9 ml

D) 27 ml

33. **Statement-1 :** Work done (W) by the system during isobaric reversible and irreversible processes from same initial state to same final state is equal.

Statement-2 : As the pressure is constant so $W = -P(\Delta V)$ for both the processes.

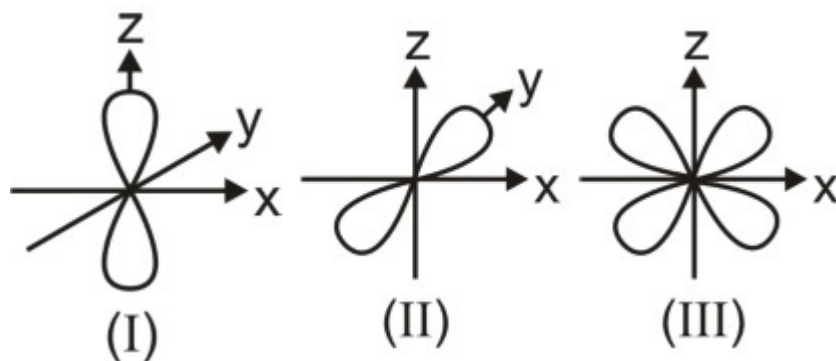
A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

C) Statement-1 is true, statement-2 is false.

D) Statement-1 is false, statement-2 is true.

34. Consider the following three orbital :



INCORRECT statement regarding above orbitals is.

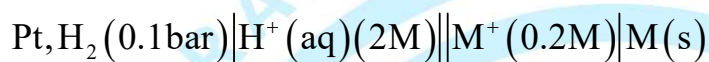
- A) If internuclear axis is 'x' then combination of (I) and (III) orbitals can form π -bond
- B) Orbital (III) can form δ -bond with other similar orbital having identical orientation of lobes if y axis is internuclear axis.
- C) If internuclear axis is 'x' then combination of (I) and (II) orbitals can form π -bond
- D) If internuclear axis is x or y or z orbitals (I) and (II) can never form any type of covalent bond.
35. The ratio of time required for an electron to complete one revolution in second orbit of H-atom and 4th orbit of He^+ ion is :
- A) 0.5 B) 0.25 C) 2 D) 4
36. $\text{A(g)} \rightleftharpoons \text{B(g)} + \text{C(g)}$ If initially A(g) is taken at 12 atm pressure in a rigid vessel at constant temperature then total pressure at equilibrium will be ?
(Given : $\Delta G^0 [\text{A(g)}] = 150 \text{ kJ/mole}$, $\Delta G^0 [\text{B(g)}] = 60 \text{ kJ/mole}$, $\Delta G^0 [\text{C(g)}] = 90 \text{ kJ/mole}$)
- A) 12 atm B) 15 atm C) 18 atm D) 9 atm
37. Consider the following hydrolysis reactions :
- (i) $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{W} + 3\text{HCl}$
- (ii) $\text{SF}_4 + 3\text{H}_2\text{O} \rightarrow \text{X} + 4\text{HF}$
- (iii) $\text{BCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{Y} + 3\text{HCl}$
- (iv) $\text{XeF}_6 + 3\text{H}_2\text{O} \rightarrow \text{Z} + 6\text{HF}$

Then according to given information the

INCORRECT statement for the given halide compounds is :

- A) In the product of (iv) reaction the hybridisation of 18th group element does not change
- B) In the product of (iii) reaction the hybridisation of 13th group element does not change
- C) In the product of (ii) reaction the hybridisation of 16th group element has been changed
- D) In the product of (i) reaction the hybridisation of 15th group element does not change

38. A Galvanic cell is represented as :



[Given : $K_w = 10^{-14}$, $E_{\text{M}^+/\text{M}}^0 = 0.03 \text{ V}$]

The correct statement for the given Galvanic cell is :

- A) $\Delta G_{\text{reaction}}^0 < 0, E_{\text{cell}} > 0$
- B) $\Delta G_{\text{reaction}}^0 > 0, E_{\text{cell}} < 0$
- C) $\Delta G_{\text{reaction}}^0 < 0, E_{\text{cell}} < 0$
- D) $\Delta G_{\text{reaction}}^0 > 0, E_{\text{cell}} < 0$

39. For a strong electrolyte AB_2 at 25°C at 0.25 M concentration, its molar conductance is 260 S-cm²/mole and at 1.0 M concentration its equivalent conductance is 125 S-cm²/eq. Its equivalent conductance at 0.04 M concentration will be ?
(in S-cm²/eq.)

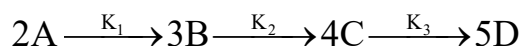
- A) 133
- B) 266
- C) 63.5
- D) 280

40. **Assertion (A) :** There is large difference between melting and boiling points of oxygen and sulphur.

Reason (R) : Strong hydrogen bonding present in H_2O which is not found in H_2S

- A) Both A and R are true but R is not the correct explanation of A
- B) A is false but R is true
- C) A is true but R is false
- D) Both A and R are true and R is the correct explanation of A

41. For the first ordered sequential reactions shown below :



(Given : K_1, K_2 and K_3 are first order rate constant of given reactions), the only incorrect option is

A) $\frac{d[A]}{dt} = -2K_1[A]$

B) $\frac{d[B]}{dt} = 2K_1[A] - 3K_2[B]$

C) $\frac{d[C]}{dt} = 4K_2[B] - 4K_3[C]$

D) $\frac{d[D]}{dt} = 5K_3[C]$

42. Match the following:

P)	Substitution reaction	i)	$CH_3 - CH_2 - Cl + alc\ KOH \rightarrow CH_2 = CH_2$
Q)	Addition reaction	ii)	$CH_3 - CH_2 - Cl + alc\ KOH \rightarrow CH_3 - CH_2 - OH$
R)	Elimination reaction	iii)	$S^- - C \equiv N$
S)	Ambident nucleophile	iv)	$CH_2 = CH_2 + HBr \rightarrow CH_3 - CH_2 - Br$

A) $P \rightarrow ii; Q \rightarrow iv; R \rightarrow i; S \rightarrow iii$

B) $P \rightarrow iii; Q \rightarrow i; R \rightarrow ii; S \rightarrow iv$

C) $P \rightarrow i; Q \rightarrow ii; R \rightarrow iii; S \rightarrow iv$

D) $P \rightarrow iii; Q \rightarrow i; R \rightarrow iv; S \rightarrow ii$

43. Select the correct statement(s).

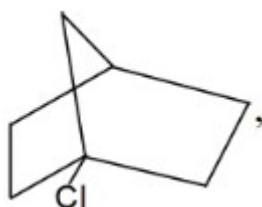
A) Stability order of complexes is $[Cr(ox)_3]^{3-} < [Cr(H_2O)_6]^{3+} < [CrF_6]^{3-}$

B) The octahedral complex formed by joining the CN^- ligands to Fe^{+3} ion has theoretical value of spin magnetic moment equal to 1.73 B.M.

C) In $CuSO_4 \cdot 5H_2O$ both Cu-O and Cu-S coordinate bonds are present

D) Stability of complex is : $[Ni(NH_3)_6]^{2+} > [Ni(en)_3]^{2+}$

44. For the given compound



the nucleophilic substitution by C_2H_5OH occurs by

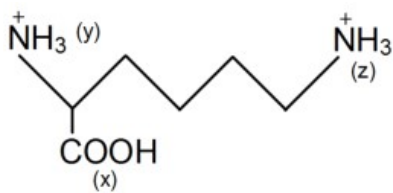
A) S_N1 mechanism

B) S_N2 mechanism

C) S_N1, S_N2 both

D) No reaction will occur

45. Acidic order of



is

- A) $(z) > (y) > (x)$ B) $(y) > (x) > (z)$ C) $(x) > (y) > (z)$ D) $(x) > (z) > (y)$

46. **Statement-1** : Mo(VI) and W(VI) are more stable than Cr (VI)

Statement-2 : In d-block elements their oxidation state can be differ from each other by unity.

Statement-3 : Scandium does not exhibit variable oxidation state

Which statement(s) is/are correct.

- A) Statement-1 & Statement-2 only
B) Statement-2 & Statement-3 only
C) Statement-1 & Statement-3 only
D) Statement-1, Statement-2 & Statement-3 all are correct

47. **Statement-1**: Ammonolysis of alkyl halides is not a suitable method for the preparation of pure primary amines.

Statement-2: Ammonolysis of alkyl halides yields mainly secondary amines.

- A) If both the statements are true and statement-2 is the correct explanation of statement-1.
B) If both the statements are true but statement-2 is not the correct explanation of statement-1.
C) If statement-1 is True and statement-2 is False.
D) If statement-1 is False and statement-2 is True.

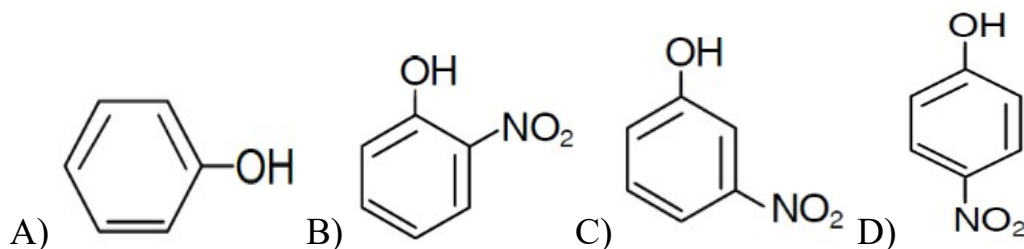
48. Which of the ester shown, after reduction with LiAlH_4 and aqueous work up, will yield two molecules of only a single alcohol ?

- A) $\text{CH}_3\text{CH}_2\text{CO}_2\text{C}_6\text{H}_5$ B) $\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_5$
C) $\text{C}_6\text{H}_5 - \text{CO}_2 - \text{CH}_2\text{C}_6\text{H}_5$ D) $\text{HCO}_2\text{C}_2\text{H}_5$

49. Yellow precipitate which is soluble in excess & hot NaOH solution.

- A) AgI B) PbBr₂ C) CdS D) PbCrO₄

50. Which of the following compound is most acidic?



SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

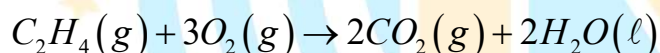
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. 400 ml 0.05 M of an aqueous solution of CH₃COOH ($K_a = 10^{-5}$) is treated with 600 ml 0.05 M of an aqueous solution of NaOH. If final concentration of CH₃COOH in the solution is $[P \times 10^{-Q}]$ then the value of $[P + Q]$ is :

[Given : $K_a(CH_3COOH) = 10^{-5}$]

52. The thermochemical equation for the combustion of ethylene gas, C₂H₄, is



$$\Delta H^\circ = -1410 \text{ kJ (at } 20^\circ\text{C)}$$

Assuming 70% efficiency during a process of combustion of ethylene gas, calculate how many kilograms of water at 20°C can be converted into steam at 100°C by burning 44.6 moles of C₂H₄ gas in standard conditions ?

[Heat capacity water liquid is 4.184 kJ/kg-K, $\Delta H_{\text{vapourization}} H_2O(l) = 2259 \text{ kJ / Kg}$]

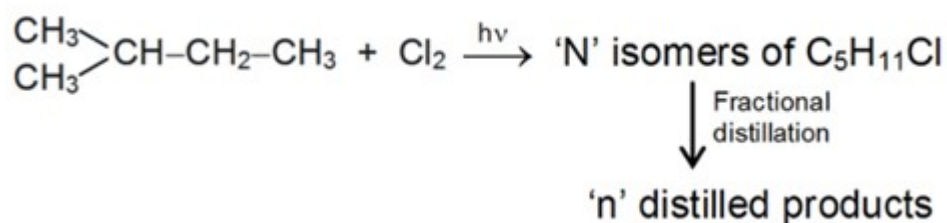
53. Among the following elements how many of them are inner transition elements?

- (a) Sg (b) Bk (c) Er (d) Sm (e) Fe (f) Pb
(g) Cr (h) Ca (i) Ar (j) Zr (k) Ce

54. Calculate the standard boiling point of $\text{PCl}_3 (\ell)$ from the given thermochemical data

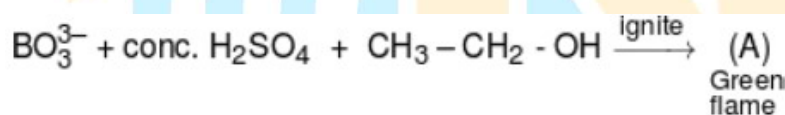
Species	Standard absolute molar entropy (S_m^0)	Standard enthalpy of formation (ΔH_f^0)
$\text{PCl}_3 (\ell)$	$217.1 \text{ JK}^{-1} \text{ mol}^{-1}$	$-319.7 \text{ kJ mol}^{-1}$
$\text{PCl}_3 (\text{g})$	$311.7 \text{ JK}^{-1} \text{ mol}^{-1}$	$-287.0 \text{ kJ mol}^{-1}$

55.

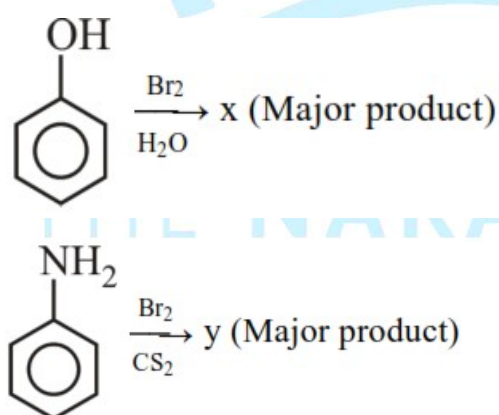


Value of $\frac{N+n}{2}$ is _____

56. Calculate number of carbon atoms in molecule of compound 'A' that is responsible from green flame.



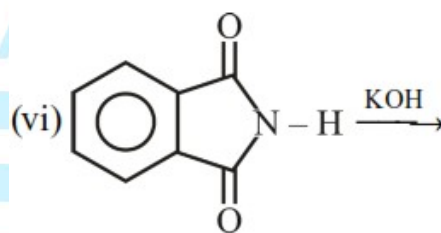
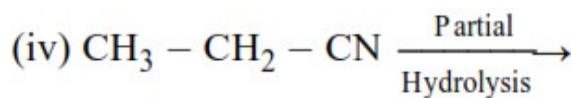
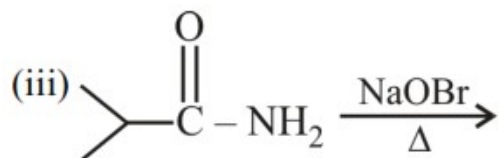
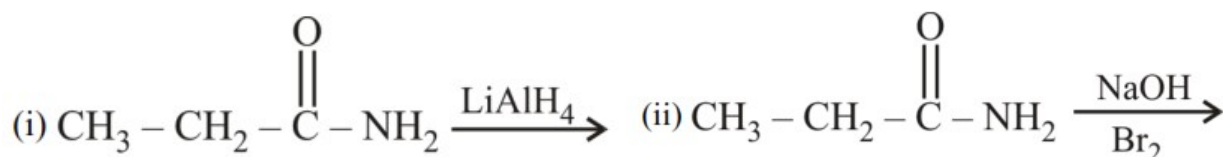
57.



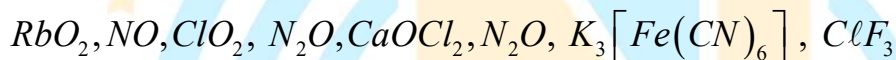
If M is number of bromine atoms per molecule of Major product x and N is number of bromine atoms per molecule of major product y. Then find out value

of $\frac{x}{y}$

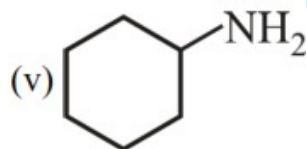
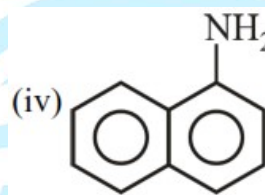
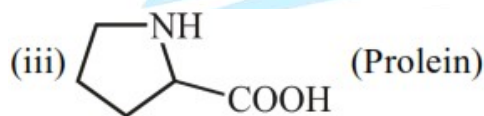
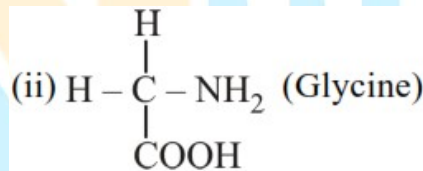
58. Correct number of reactions have 1° amine is major product:



59. How many of the following compounds are having odd number of electrons.



60. Number of compounds show carbylamine test :



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Number of words which can be formed using all the letters of the word 'HIP HIP HURRAY' in which all H's lie somewhere between R's is
- A) $(198)7!$ B) $(99)7!$ C) $(99)8!$ D) $(198)8!$
62. If $f(x) = \operatorname{sgn}(3x \cos^{-1} x - 6 \cos^{-1} x - \pi x + 2\pi)$, then the number of points of discontinuity of $f(x)$ is/are
- A) 2 B) 0 C) 1 D) 3
63. **Assertion (A) :** Slope of the curve given as $y = x^2$ at $x = 1$ is not defined.
- Reason (R) :** Slope of the curve given as $y = x^2$ at $x = 1$ is $\pm \frac{1}{2}$
- A) Both A and R are true and R is the correct explanation of A
B) Both A and R are true but R is NOT the correct explanation of A
C) A is true but R is false.
D) Both A and R are false.
64. Given $\vec{a}, \vec{b}, \vec{c}$, are vectors such that $[\vec{a} \vec{b} \vec{c}] = \frac{1}{3}$. If the vector $\vec{V}$ can be expressed as linear combination of $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$ as $\vec{V} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$ then $(x + y + z)$ has the value equal to
- A) $\vec{V} \cdot (\vec{a} + \vec{b} + \vec{c})$ B) $3\vec{V} \cdot (\vec{a} + \vec{b} + \vec{c})$ C) $2\vec{V} \cdot (\vec{a} + \vec{b} + \vec{c})$ D) None
65. The number of 12^{th} roots of unity which are also the 18 root of unity is/are
- A) 1 B) 2 C) 3 D) 6
66. If cofactors of elements of a diagonal matrix A of order 3 are the roots of the equation $x^9 + kx^8 - 16x^6 = 0, k \in \mathbb{R}$ then $|\det(A)|$ is equal to
- A) 4 B) 8 C) 16 D) 256

67. All possible 5 lettered words are formed using all the letters of the word "COCOA" and a word from it is randomly selected. The chance that any two alike letters are not adjacent in the selected word, is

- A) $\frac{1}{5}$ B) $\frac{2}{5}$ C) $\frac{1}{3}$ D) $\frac{2}{3}$

68. Area of the triangle formed by tangent of the curve $x=t+t^2+2$, $y=t^2-t+2$, $t \geq 0$ which passes through (2, 2) and co-ordinate axes is equal to

- A) 2 B) 4 C) 6 D) 8

69. Words are formed using all the letters of the word 'BANKEBIHARI'. Match the following.

LIST-I		LIST-II	
A)	Number of words which contain the word 'KHER' and no two vowels occur together, is	I)	114
B)	Number of words which contain the word 'BIKANER' is	II)	5!
C)	Number of words which contain the word 'BIKANER' but not the word 'BIKANERIA' is	III)	$3 \times 5!$
D)	Total number of words using all letters of word 'BANKEBIHARI'	IV)	$\frac{11!}{(2!)^3}$

A) A $\rightarrow$ III; B $\rightarrow$ II; C $\rightarrow$ I; D $\rightarrow$ IV B) A $\rightarrow$ IV; B $\rightarrow$ III; C $\rightarrow$ II; D $\rightarrow$ I

C) A $\rightarrow$ I; B $\rightarrow$ II; C $\rightarrow$ III; D $\rightarrow$ IV D) A $\rightarrow$ II; B $\rightarrow$ III; C $\rightarrow$ IV; D $\rightarrow$ I

70. The value of definite integral $\int_{-5}^2 \frac{3x^2 + x - 3}{3x^2 + 9x + 9} dx$ is equal to

- A) 3 B) 5 C) 7 D) 19

71. If tangents are drawn from any point on the curve $\frac{x-1}{\cos \phi} = \frac{y-2}{\sin \phi} = 6$ to the curve

$\frac{x-1}{\cos \theta} = \frac{y-2}{\sin \theta} = 3$ where θ and ϕ are parameters then acute angle between tangents is

- A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{2\pi}{3}$ D) $\frac{5\pi}{6}$

72. If the point representing the complex number z_k is lying on or inside the circle having centre $(1 + 0i)$ and radius is k , then the maximum value of $|z_1 + z_2 + z_3 + z_4 + \dots + z_{20}|$ is
- A) 210 B) 220 C) 230 D) 260
73. Let R be the region satisfying $y < x^2 + 1$, $y > x - 1$, $x < 1$ and $x > 0$, then area of R is
- A) $\frac{11}{6}$ B) $\frac{3}{2}$ C) $\frac{5}{6}$ D) 2
74. The value of $\int_0^{\pi/2} \frac{dx}{(\sqrt{\cos x} + \sqrt{\sin x})^4}$ is equal to
- A) 3 B) $\frac{2}{\sqrt{3}}$ C) $\frac{2}{3}$ D) $\frac{1}{3}$
75. Consider 3 equal circles of radius r , within a circle of radius r_2 , each to touch the other two and the given circle.
- Statement-1: $\frac{r}{r_2} = \frac{\sqrt{3}}{\sqrt{3} + 1}$
- Statement-2: Incentre of triangle formed by joining centres of 3 equals is same as centre of given circle.
- A) Statement-1 is true, Statement-2 is true.
B) Statement-1 is false, Statement-2 is false.
C) Statement-1 is true, Statement-2 is false.
D) Statement-1 is false, Statement-2 is true.
76. Let $f(x) = |x - 1| + (x - 2)$, $I = \int_0^3 f(x) dx$, $M =$ then minimum value of f , $N = f'(x)$ for $x < -4$ and $C =$ the value of $f''(4)$. Then the value of $\frac{M^2 - N^2 + IC}{2}$ is
- A) $-\frac{3}{2}$ B) $-\frac{5}{2}$ C) $\frac{3}{2}$ D) $\frac{5}{2}$

77. Let $f(x) (x \geq 1)$ be a differentiable function satisfying $f(x) = (\log x)^2 - \int_1^x \frac{f(t)}{t} dt$.

Area bounded by tangent line of $y = f(x)$ at the point $(e, f(e))$, the curve $y = f(x)$ and the line $x = 1$, is

- A) $e + \frac{1}{e}$ B) $e + \frac{1}{e} - 1$ C) $e + \frac{1}{e} - 2$ D) $e + \frac{1}{e} - 3$

78. If $A = \begin{pmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{pmatrix}$ and $B = \begin{pmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix}$ the value of determinant

$(AB^2 + A^2B^4 + A^3B^6 + \dots + 101 \text{ terms})$ is

- A) 0 B) 101 C) 5049 D) 5050

79. The value of $\int_{\pi/2}^{5\pi/2} \frac{e^{\tan^{-1}(\sin x)}}{e^{\tan^{-1}(\sin x)} + e^{\tan^{-1}(\cos x)}} dx$ is equal to

- A) $\frac{\pi}{2}$ B) 2π C) π D) $\frac{\pi}{4}$

70. The distance of the point $P(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is

- A) 1 B) 2 C) $\sqrt{5}$ D) 5

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If $\alpha \neq 1$ is the seventh root of unity then the value of

$$\left\lfloor \log_{\sqrt{7}} \left| 1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 - \frac{13}{\alpha} \right| \right\rfloor$$

[Note : $[y]$ denotes greatest integer function less than or equal to y .]

82. If $\lim_{x \rightarrow 0} \frac{1 + 3a \cos 2x + 5b \cos 4x}{x^4}$ exists for all $x \in \mathbb{R}$ and has value 'c' then find the value of $[a - b + c]$.

[Note: $[y]$ denotes greatest integer function less than or equal to y .]

83. In an increasing G.P., the sum of the first and the last term is 66, the product of the second and the second last term is 128, and the sum of all the terms is 126. How many terms are there in the progression ?
84. A normal is drawn at a point $P(x, y)$ on a curve. It meets the x -axis and the y -axis at A and B respectively such that $\frac{1}{OA} + \frac{1}{OB} = 1$, where O is origin, then find radius of the director circle of the curve passing through $(3, 3)$.
85. Four people sit around a circular table, and each person will roll a normal six sided die once. The probability that no two people sitting next to each other will roll the same number is $\frac{N}{1296}$. Then the value of N .
86. Given $2x - y + 2z = 1$; $x - 2y + z = -4$ & $x + y + \lambda z = 4$. Then the value of λ such that the given system of equation has no solution is
87. If for non-zero x , $3f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} - 10$, then $\int_2^3 f(x) dx = \frac{a}{b} \ln\left(\frac{c}{a}\right)$ where a, b, c are co-prime then the value of $(a + b + c)$
88. Let z_1, z_2, z_3 be complex numbers such that $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$ and $|z_1 + z_2 + z_3| = 21$. Given that $|z_1 - z_2| = 2\sqrt{3}, |z_1| = 3\sqrt{3}$. Then $|z_2|^2 + |z_3|^2$ is
89. If angle between two focal chords of a parabola $(y - 5)^2 = 8(x - 1)$ which are tangents to the circle $x^2 + y^2 = 9$ is $\tan^{-1}\left(\frac{a}{b}\right)$, where a and b relatively prime number then the value of $a + b$.
90. If α and β are the roots of the equation $x^2 - 6x + 12 = 0$ and the value of $(\alpha - 2)^{24} - \frac{(\beta - 6)^8}{\alpha^8} + 1$ is a^b where a and b are natural numbers then least value of $(a + b)$ is